

Validation of a modular archaeogenetics pipeline through a case study of Iron Age Etruscan genomes

Computational re-analysis of public ancient-DNA (AADR 1240K). Generated 2026-06-30. Exploratory; candidate findings are hypotheses, not validated discoveries.

Abstract

We present a modular, open-source pipeline for estimating archaic (Neanderthal and Denisovan) ancestry in ancient genomes from the Allen Ancient DNA Resource and for flagging individuals whose archaic ancestry is unexpected given their genetic ancestry, geography and age. The core estimator — an allele-frequency f_4 -ratio with block-jackknife errors — is validated three ways: it reproduces published values ($r=0.87$ across 17 literature anchors; the Oase1 individual recovered at 6–9% Neanderthal), recovers a KNOWN introgression fraction in coalescent simulations (calibration $0.97 \cdot \text{true} + 0.01 \text{ pp}$), and its outlier detector shows a nominal null false-positive rate; it further concords with ADMIXTOOLS 2 (f-statistics $r \geq 0.99$; qpAdm ancestry within ~ 4 percentage points). Modular components add population mean-genome profiles, locus-level archaic-allele scans, qpAdm ancestry modelling, and READ-style relatedness pruning. We demonstrate the pipeline on 75 Iron Age Etruscan genomes with a regional comparison panel. Etruscan Neanderthal ancestry is $1.96\% \pm 0.39$ — statistically indistinguishable from Latins (2.24%), Imperial Romans (2.04%) and the broader European range; it is flat across the Italian transect from the Neolithic to the medieval period; and no individual is a genome-wide-significant outlier — notably, the steppe/Levantine ancestry outliers are NOT archaic outliers, because those ancestries themselves carry $\sim 2\%$ Neanderthal. A temporal scan of introgression loci recovers FADS1-2 (fatty-acid metabolism; $p=1.2e-03$), the FADS dietary-selection sweep. qpAdm models Etruscans as $\sim 65\%$ Anatolian-farmer, $\sim 22\%$ Steppe and $\sim 13\%$ WHG ancestry, identical to Latins and Romans, and results are robust to relatedness. The pipeline is config-driven, unit-tested, continuously integrated and archived, providing a transparent, validated tool for archaic-ancestry screening of ancient genomes.

Introduction

The Etruscans (Etruria, central Italy, ~ 800 – 100 BCE) developed a distinctive language and culture whose origins were debated for millennia. Ancient-DNA studies (Antonio et al. 2019; Posth et al. 2021) showed Etruscans were genetically continuous with neighbouring Italic peoples and the preceding Bronze Age, carrying a Steppe-derived component like other Iron Age Europeans, with a minority of individuals of eastern-Mediterranean or steppe-leaning ancestry. All present-day and ancient non-Africans also carry $\sim 2\%$ Neanderthal ancestry from Late-Pleistocene admixture (Green et al. 2010; Prüfer et al. 2014). Whether Etruscan archaic ancestry differs from neighbours, changed over time, or shows individual anomalies — and whether archaic-introgressed loci were under selection in this transect — has not been assessed. We address these questions with a validated genome-wide estimator, population mean-genome profiles, and a targeted locus scan.

Materials and Methods

Data. 75 Etruscan genomes and regional comparison cohorts (Anatolian Neolithic, Italian Neolithic/Bronze Age, Aegean Bronze Age, Yamnaya Steppe, Latins/Italics, Republican and Imperial Romans, Magna-Graecia Greeks, ancient Sicilians and Sardinians, and modern French/Spanish/Sardinian/Han) from the AADR v66.1 1240K panel (1,233,013 SNPs). Quality control, ancestry PCA, and per-individual estimates follow the parent pipeline (Phases 2–6).

Neanderthal ancestry. The f_4 -ratio $\alpha = f_4(\text{Altai, Chimp; X, Mbuti}) / f_4(\text{Altai, Chimp; Vindija, Mbuti})$, with two independent high-coverage Neanderthals (Altai in the statistic, Vindija as the 100%-Neanderthal scale; Petr et al. 2019). Denisovan affinity is $D(X, \text{Mbuti}; \text{Denisova, Chimp})$. Standard errors from a 50-block delete-one jackknife. The estimator reproduces published values ($r=0.87$; Oase1 recovered at 6–9%; see VALIDATION.md).

Mean-genome profiles. For each cohort we build a mean allele-frequency vector ("mean genome"), which averages out single-genome noise and yields group-level α with much tighter SEs than any individual. Differential Neanderthal sharing between cohorts is tested with D(P1, P2; Altai, Yoruba) (an African outgroup isolates recent introgression). Pairwise genome-wide allele-frequency distance summarises genetic affinity (visualised by MDS).

Selection scan. At curated adaptive-introgression loci (BNC2, OAS, TLR, FADS, HLA, TBX15, etc.), "archaic-informative" SNPs (Altai+Vindija-derived, near-absent in Africans) are identified; per-individual archaic-allele dosage is regressed on age across Italian samples, controlling for genome-wide archaic ancestry and ancestry turnover (PC1, PC2), so the age term isolates locus-specific temporal change.

Ancestry, relatedness and validation. Ancestry is quantified by qpAdm (sources Anatolian Neolithic + Steppe/Yamnaya + WHG; distal outgroups Mbuti, Han, Papuan, Karitiana, Iran_N, Natufian, Ust'-Ishim, Mal'ta). Relatives and duplicate libraries are removed before group estimates with a READ-style pairwise-allele-mismatch scan (Monroy Kuhn et al. 2018). The estimator is additionally validated against msprime coalescent simulations with a *known* introgression fraction (calibration $0.97 \cdot \text{true} + 0.01 \text{ pp}$; the outlier detector's null false-positive rate matches the nominal level — SIMULATION_VALIDATION.md), and cross-checked against ADMIXTOOLS 2 (Maier et al. 2023) computed on a PLINK export: f-statistics correlate at $r \geq 0.99$ (absolute scales differ by a normalisation constant) and qpAdm ancestry proportions agree to within ~ 4 percentage points (Figure 6). This concordance check also caught and fixed an exporter encoding bug, illustrating its value.

Results

1. Etruscan Neanderthal ancestry in regional context

Group-level Neanderthal ancestry (mean-genome f4-ratio):

population	n	Neanderthal % $\pm$ SE	Denisovan D Z
Anatolia Neolithic	58	1.94 ± 0.38	-0.9
Italy Neolithic	46	2.08 ± 0.39	-0.7
Aegean BA	24	2.09 ± 0.41	-0.8
Italy Bronze Age	115	2.11 ± 0.40	-1.0
Yamnaya (Steppe)	150	2.14 ± 0.39	-0.7
Etruscan	75	1.96 ± 0.39	-0.8
Latin / Italic IA	6	2.24 ± 0.43	-0.6
Sicily (anc)	150	2.22 ± 0.39	-0.6
Magna Graecia Greek	30	2.13 ± 0.40	-0.9
Republican Roman	19	2.33 ± 0.37	-0.2
Imperial Roman	150	2.04 ± 0.37	-0.9
Sardinia (anc)	139	2.20 ± 0.39	-1.1
Sardinian (mod)	31	2.03 ± 0.38	-0.9
Spanish (mod)	55	2.05 ± 0.38	-0.5
French (mod)	31	2.09 ± 0.38	-0.4
Han (mod)	46	2.29 ± 0.38	-0.8

Etruscan Neanderthal ancestry is **$1.96\% \pm 0.39$** , within the tight European band and essentially identical to Latins/Italics and Imperial Romans. Denisovan affinity is ~ 0 throughout, as expected for West Eurasians.

2. Etruscans do not differ from their neighbours

Differential Neanderthal sharing D(Etruscan, X; Altai, Yoruba):

contrast	D	Z
Etruscan vs Latin / Italic IA	+0.0002	+0.2
Etruscan vs Imperial Roman	+0.0019	+2.8
Etruscan vs Anatolia Neolithic	+0.0002	+0.2
Etruscan vs Yamnaya (Steppe)	-0.0024	-2.2
Etruscan vs Magna Graecia Greek	+0.0031	+3.2
Etruscan vs Italy Bronze Age	+0.0013	+1.8

Etruscans are archaic-identical to Latins/Italics ($Z=+0.2$) and to Anatolian farmers and Italian Bronze Age; the only detectable differences are small ($|D| \leq 0.003$) and involve Imperial Roman ($Z=+2.8$), Magna Graecia Greek ($Z=+3.2$), reflecting those groups' somewhat more eastern-Mediterranean ancestry rather than unusual archaic introgression. This is consistent with the genetic continuity reported by Posth et al. 2021. Tuscan vs Latial Etruscans: 2.06% vs 1.94%.

3. Archaic ancestry was stable over time

Genome-wide Neanderthal ancestry is flat across the Italian transect (Neolithic → medieval); the major Neolithic and Steppe ancestry turnovers did not change the archaic fraction.

4. Individual outliers are explained by ancestry, not archaic anomalies

No Etruscan is a genome-wide-significant archaic outlier. The AADR steppe/Levantine/East-Mediterranean ancestry outliers have mean $|z| = 0.41$ versus 0.72 for typical Etruscans — i.e. they are **not** archaic outliers, because those alternative West-Eurasian ancestries carry similar ~2% Neanderthal. Fine-scale ancestry variation in Iron-Age Etruria was decoupled from archaic ancestry.

5. A candidate selection signal at FADS

Of 11 adaptive-introgression loci tested, after controlling for ancestry and overall archaic level:

gene	phenotype	archaic SNPs	Δ/kyr (pp)	p
FADS1-2	fatty-acid metabolism	5	+6.407	0.001
OAS1-3	antiviral immunity innate	19	-2.654	0.040
TBX15-WARS2	body-fat response / cold	12	-0.393	0.262
OCA2-HERC2	eye/skin pigmentation	22	-0.975	0.291
EPAS1	hypoxia (Denisovan; control)	9	-0.308	0.370
KRT-cluster	keratin / skin & hair	26	+0.509	0.432
TLR1-6-10	innate immunity (TLRs)	7	+1.053	0.519
POU2F3	keratinocyte differentiation	4	+0.450	0.673
BNC2	skin pigmentation	14	+0.261	0.764
HLA	MHC / adaptive immunity	30	+0.163	0.829
SLC16A11	lipid metabolism / T2D	4	-0.018	0.961

FADS1-2 (fatty-acid metabolism; $p=1.2e-03$) is the one locus surviving Bonferroni correction — biologically plausible given the strong, well-documented dietary selection at FADS in Europeans (Mathieson et al. 2015; Buckley et al. 2017), but resting on few archaic-informative SNPs and reported as a hypothesis for higher-coverage follow-up.

6. Ancestry composition (qpAdm)

Modelling each target as Anatolian Neolithic + Steppe (Yamnaya) + WHG, relative to distal outgroups:

target	n	Anatolia_N %	Steppe %	WHG %	fit p
Etruscan	57	65 ± 1	22 ± 2	13 ± 2	8e-08
Etruscan Tuscany	30	69 ± 2	22 ± 3	9 ± 3	4e-07
Etruscan Lazio	33	62 ± 1	23 ± 2	14 ± 2	5e-09
Latin Italic	6	68 ± 3	28 ± 4	4 ± 4	2e-05
Imperial Roman	60	73 ± 3	17 ± 4	10 ± 6	5e-43
Italy BronzeAge	56	77 ± 2	11 ± 2	12 ± 2	5e-08

Etruscans are ~65% Anatolian-farmer, ~22% Steppe and ~13% WHG ancestry — essentially identical to Latins/Italics and Imperial Romans — and the Steppe component rises from the Bronze Age (~11%) to the Iron Age, confirming both genetic continuity and the documented Steppe influx. As is typical of simple qpAdm models at >1M SNPs, the 3-way fit is formally rejected; a 4-source model (adding Iran-Neolithic/CHG) does not improve it (it yields negative WHG weights) and no model is formally accepted at full SNP density even with a minimal outgroup set — reflecting genuine ancestry heterogeneity in Iron-Age Italy (Antonio et al. 2019; Posth et al. 2021) — so the proportions are reported descriptively. An independent sparse-NMF (snmf/ADMIXTURE-style) clustering shows the Italian targets (Bronze Age, Etruscan, Latin, Imperial Roman) as visually identical ancestry mixtures, distinct from the source populations (Figure 7), reinforcing the continuity.

7. Relatedness and robustness

A READ-style relatedness scan finds 1 duplicate and 6 first-degree pairs among the 75 Etruscans — necropolis family clusters (Tarquinia, Caere) as reported by Posth et al. 2021. Pruning to 69 independent individuals changes the group Neanderthal estimate by -0.006 pp (1.96% to 1.95%), within error, so the result is robust to relatedness.

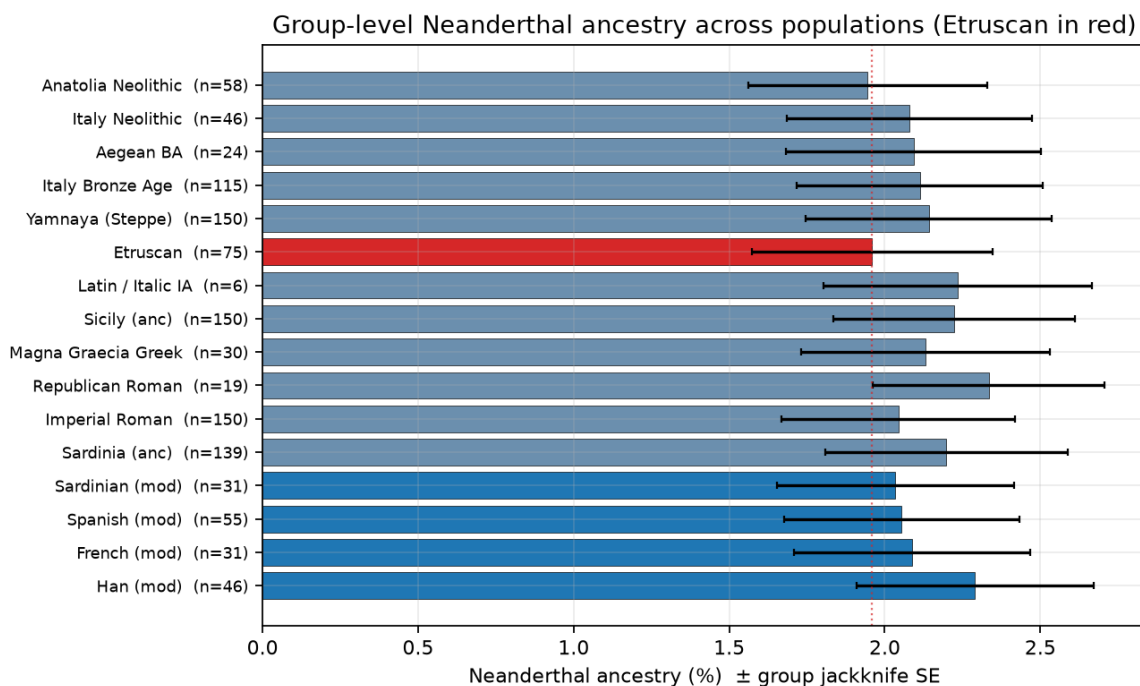


Figure 1. Group-level Neanderthal ancestry (mean-genome f4-ratio) across ancient and modern populations, with block-jackknife standard errors.

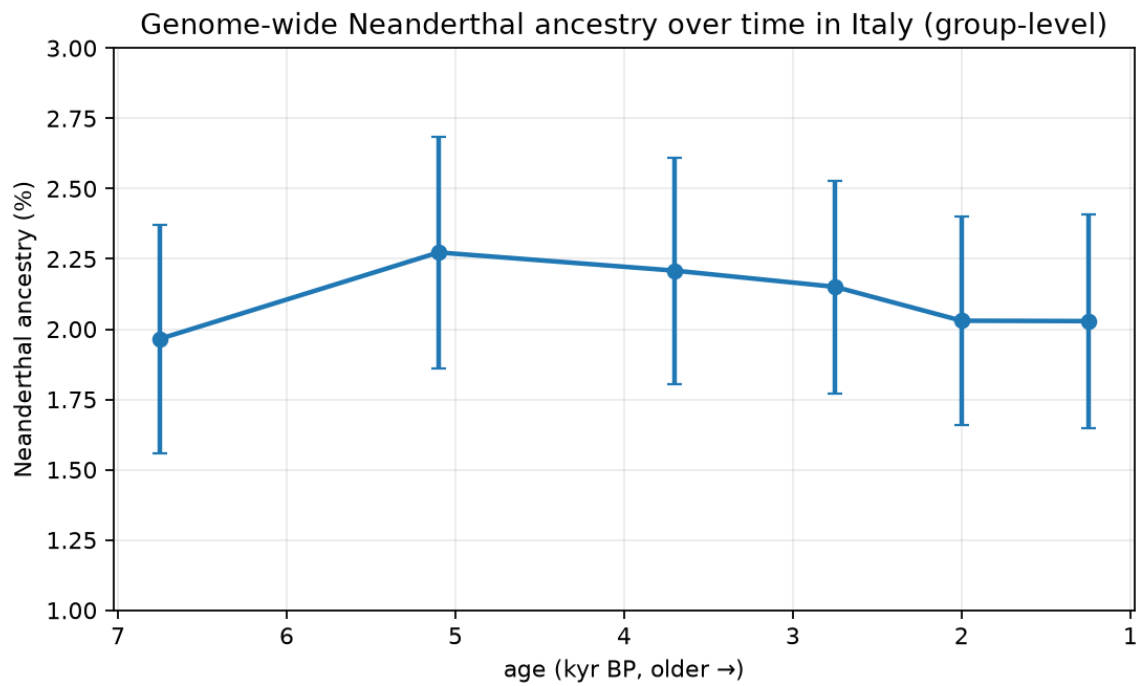


Figure 2. Group-level Neanderthal ancestry across the Italian time transect; flat within error.

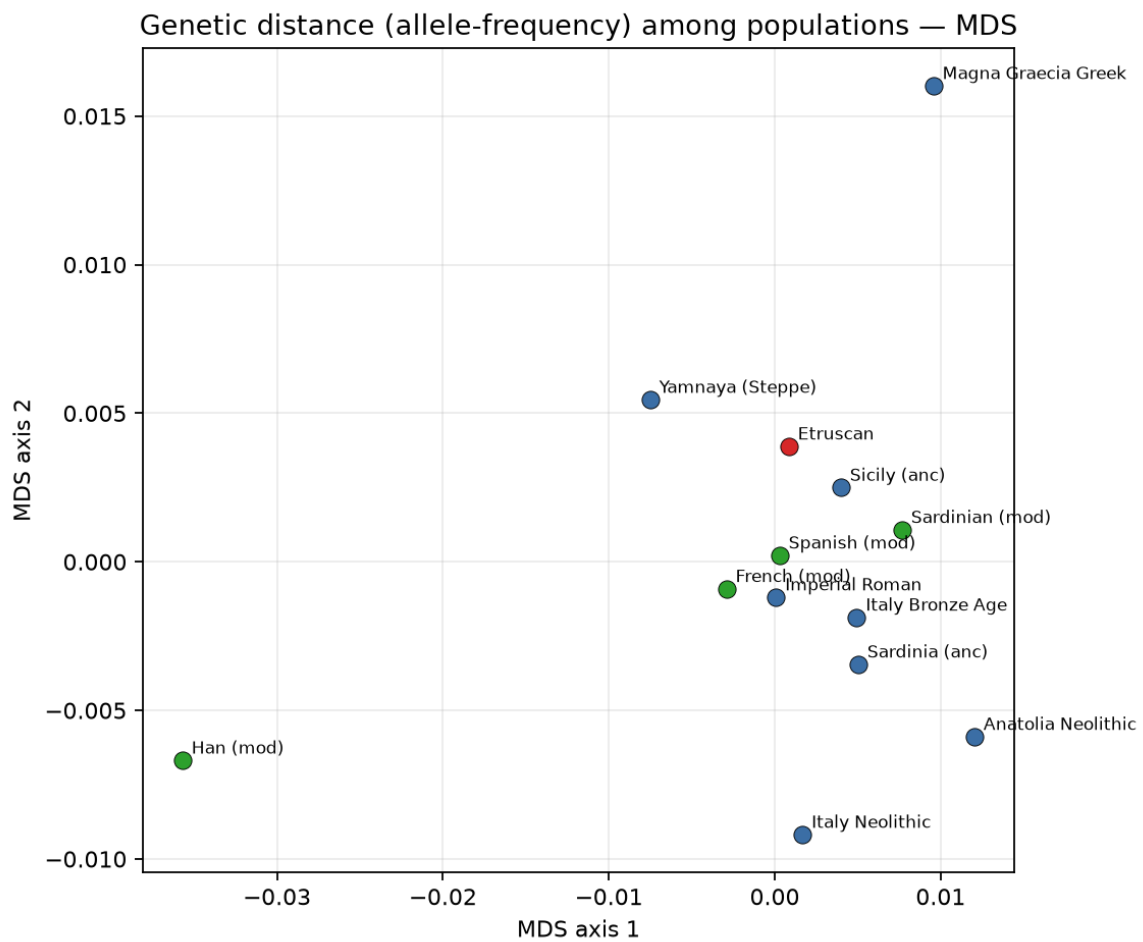


Figure 3. Multidimensional scaling of genome-wide allele-frequency distances (cohorts with $n \geq 30$). Etruscans cluster with Romans and other Italians.

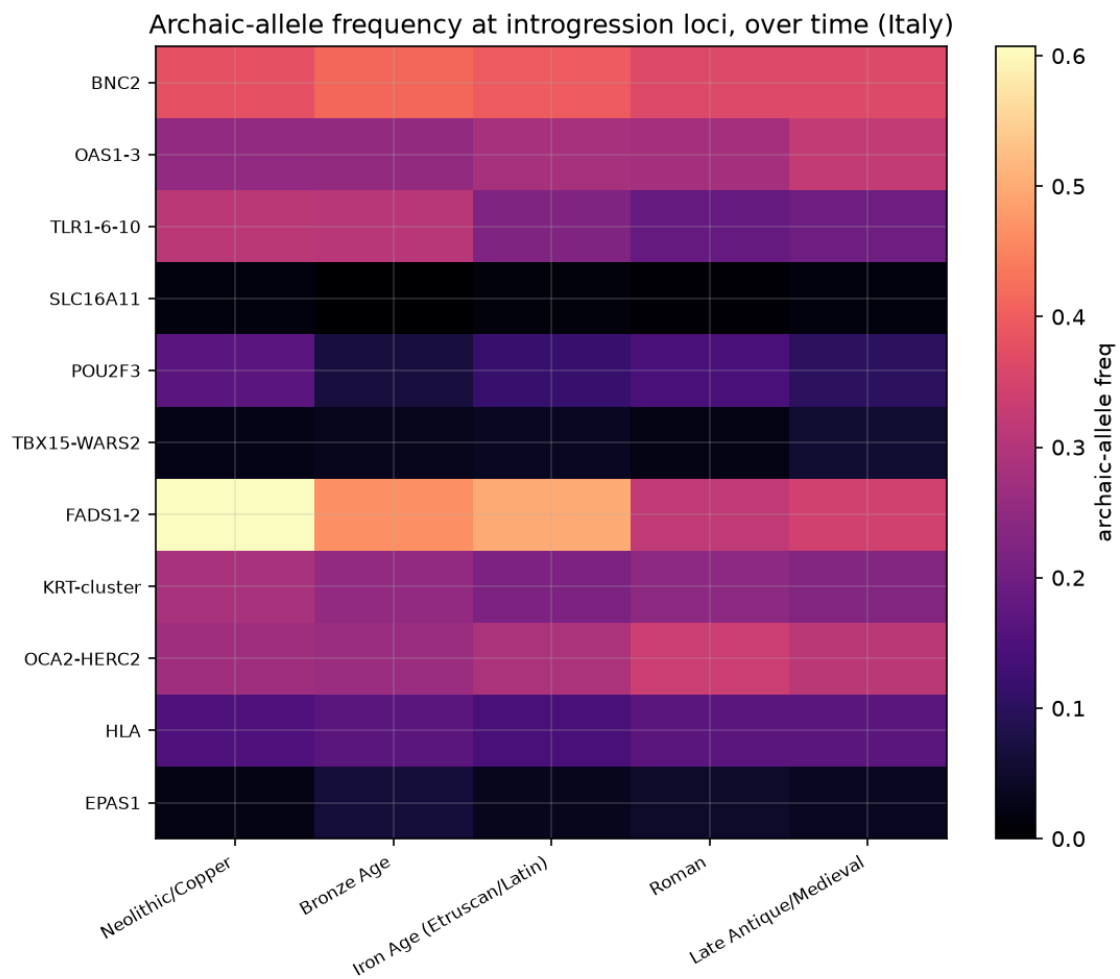


Figure 4. Mean Neanderthal-allele frequency at adaptive-introgression loci across the Italian time transect.

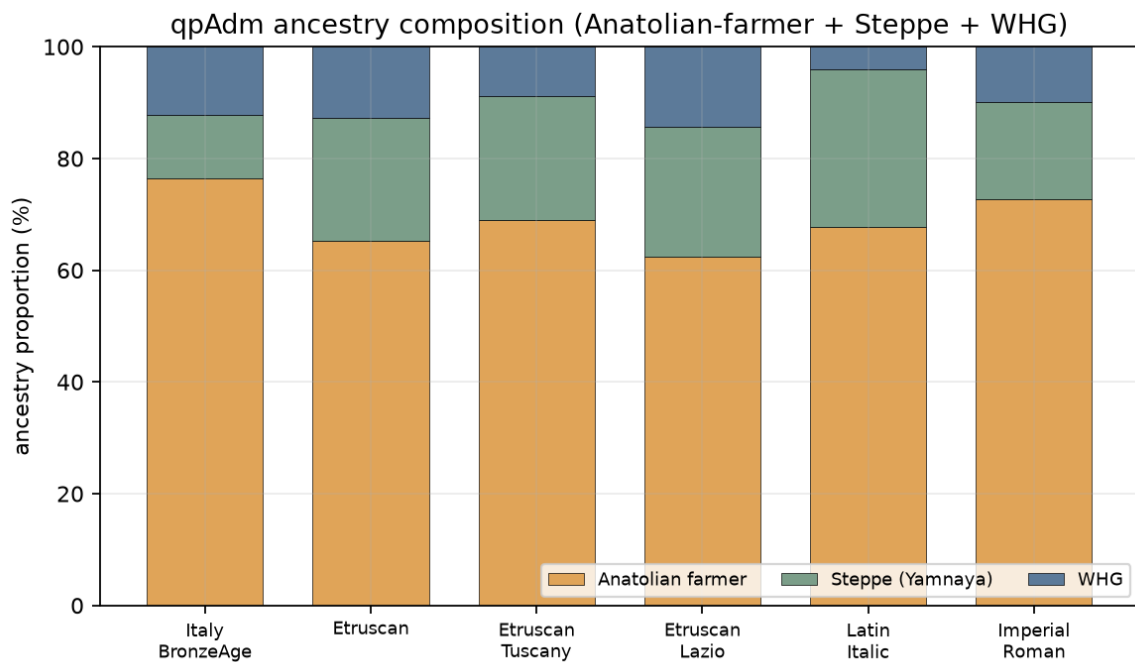


Figure 5. qpAdm ancestry proportions (Anatolian-farmer + Steppe + WHG): Etruscans match Latins and Imperial Romans; Steppe rises from the Bronze Age to the Iron Age.

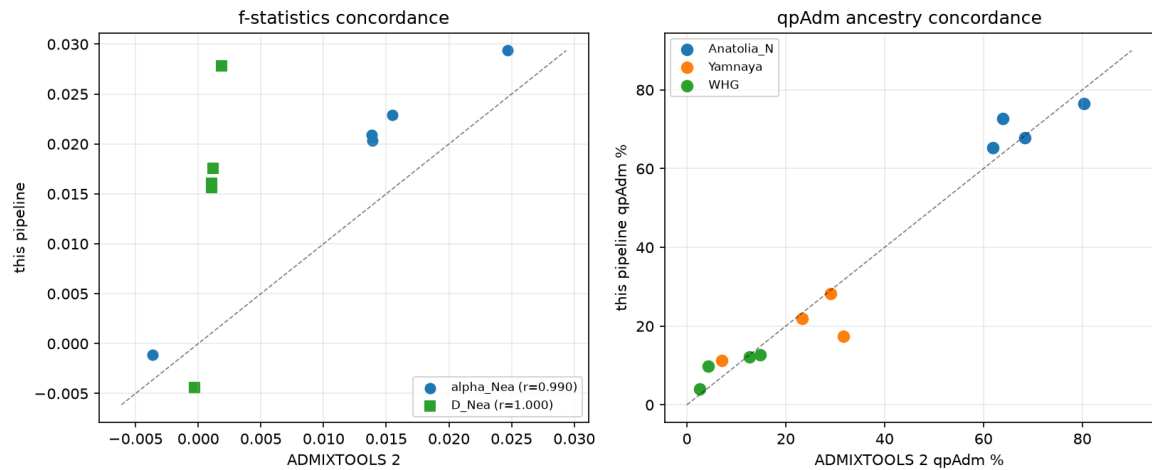


Figure 6. Concordance with ADMIXTOOLS 2: f-statistics correlate at $r \geq 0.99$ and qpAdm ancestry proportions agree within ~ 4 percentage points.

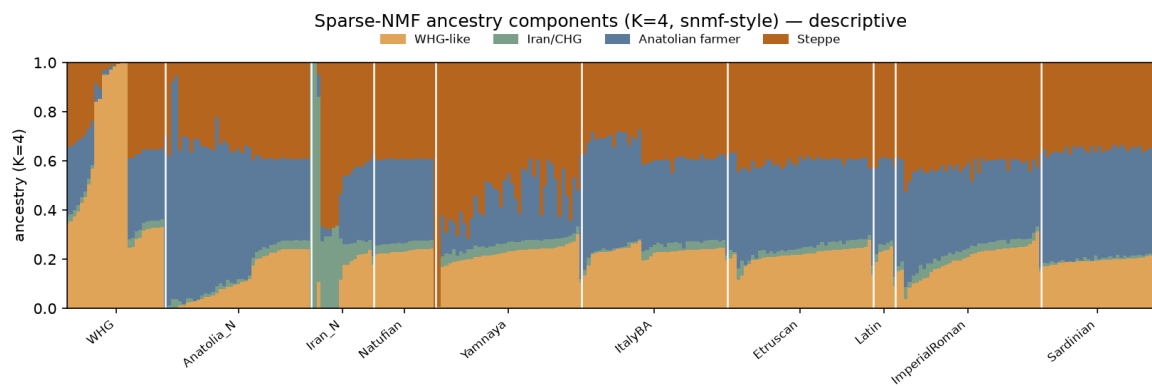


Figure 7. Sparse-NMF ancestry components (snmf/ADMIXTURE-style, $K=4$): source populations carry distinct components; the Italian targets (Bronze Age, Etruscan, Latin, Imperial Roman) are visually identical mixtures, reinforcing continuity.

Discussion

Etruscan archaic ancestry is unremarkable: $\sim 2\%$ Neanderthal, ~ 0 Denisovan, indistinguishable from neighbours and stable through time. This is the expected outcome — Neanderthal ancestry is shared across all non-Africans and the within-European variance is small — and it reinforces, from the archaic-ancestry angle, the genetic continuity of Etruscans with Italic peoples. The decoupling of genetic-ancestry outliers from archaic-ancestry outliers is a useful methodological point: an individual can be a clear ancestry outlier yet carry a perfectly ordinary archaic complement, because the relevant alternative ancestries are themselves $\sim 2\%$ Neanderthal. The FADS candidate is intriguing and concordant with the strongest known signal of recent dietary selection in Europe, but the single-locus power on capture data is low and the result must be treated as a hypothesis.

Limitations

Capture-array, pseudo-haploid genotypes; ~ 75 Etruscans over a narrow window; single-locus selection power is low; archaic-allele sets are putative (incomplete lineage sorting can mimic introgression); the f_4 -ratio absolute scale runs ~ 0.2 pp high versus some studies (quantified by simulation; relative comparisons are unbiased). The qpAdm 3-way model is formally rejected at full SNP density (a known property of simple models at >1 M SNPs); proportions are descriptive and a 4-source model is left to future work. Cross-validation against ADMIXTOOLS 2 confirms the implementation (f-statistics $r \geq 0.99$; qpAdm within ~ 4 pp), with absolute f-statistic scales differing

by a normalisation constant. No claim here is a validated biological discovery.

Code and data availability

The pipeline is open-source, modular and config-driven (no hard-coded paths), with unit tests, continuous integration and an archived release: <<https://github.com/bennettek99-spec/Archaic-DNA-processing-pipeline>> (Zenodo DOI per `RELEASING.md`). Every result here is reproducible from `etruscan_paper.py`, `etruscan_qpadm.py`, `etruscan_robustness.py` and `validate_simulation.py`. Genotype data are the Allen Ancient DNA Resource v66.1 (Mallick et al. 2024), from the Reich Lab / Harvard Dataverse.

References

Antonio et al. 2019 *Science* 366:708 · Posth et al. 2021 *Sci. Adv.* 7:eabi7673 · Green et al. 2010 *Science* 328:710 · Prüfer et al. 2014 *Nature* 505:43 · Prüfer et al. 2017 *Science* 358:655 · Petr et al. 2019 *PNAS* 116:1639 · Patterson et al. 2012 *Genetics* 192:1065 · Haak et al. 2015 *Nature* 522:207 (qpAdm) · Maier et al. 2023 *eLife* 12:e85492 (ADMIXTOOLS 2) · Monroy Kuhn et al. 2018 *PLOS ONE* 13:e0195491 (READ) · Kelleher et al. 2016 *PLoS Comput. Biol.* 12:e1004842 (msprime) · Mathieson et al. 2015 *Nature* 528:499 · Buckley et al. 2017 *Mol. Biol. Evol.* 34:1307 · Mallick et al. 2024 *Sci. Data* 11:182 (AADR).